



Sentiment Analysis

```
In [1]: # Import necessary libraries
import numpy as np
import pandas as pd

#Read the data
df=pd.read_csv("Reviews.csv",nrows=100)

#Look % Top row
df.head(5)
```

```
Out[1]:
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	Help
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian		1
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa		0
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"		1
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl		3
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"		0

```
In [2]: #looking at summary
df.Summary.head()
```

```
Out[2]: 0    Good Quality Dog Food
1    Not as Advertised
2    "Delight" says it all
3    Cough Medicine
4    Great taffy
Name: Summary, dtype: str
```

```
In [3]: #Looking at the decription of the reviews
df.Text.head()
```

```
Out[3]: 0    I have bought several of the Vitality canned d...
1    Product arrived labeled as Jumbo Salted Peanut...
2    This is a confection that has been around a fe...
3    If you are looking for the secret ingredient i...
4    Great taffy at a great price. There was a wid...
Name: Text, dtype: str
```

```
In [4]: #Import libraries
from nltk.corpus import stopwords
from textblob import TextBlob
from textblob import Word

# Lower casting and removing punctuations
```

```
In [5]: df['Text'] = df['Text'].apply(lambda x: " ".join(x.lower() for x in x.split()))
df['Text']=df['Text'].str.replace('[^\w\s]','')
import nltk
nltk.download('stopwords',quiet=True)
nltk.download('wordnet',quiet=True)
nltk.download('punkt',quiet=True)
#Removal of stop words
stop=stopwords.words('english')
df['Text']=df['Text'].apply(lambda x: " ".join(x for x in x.split() if x not i

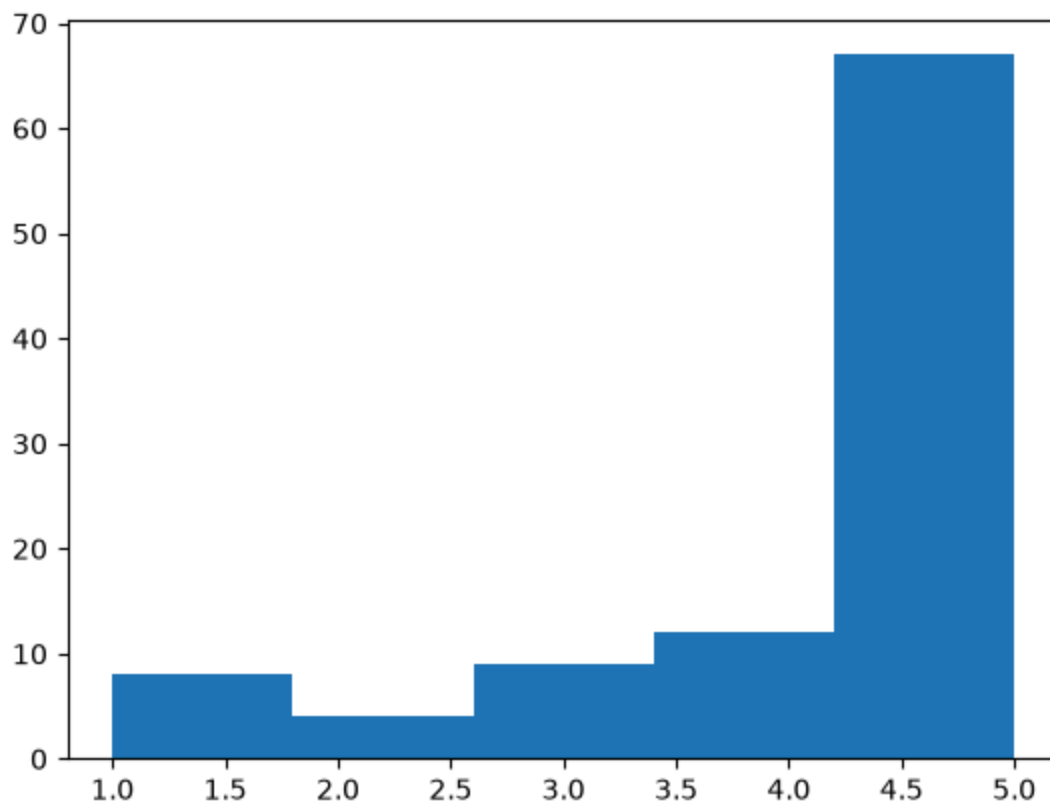
# Spelling correction
df['Text']=df['Text'].apply(lambda x: str(TextBlob(x).correct()))

#Lemmatization
df['Text']=df['Text'].apply(lambda x: " ".join([Word(word).lemmatize() for wor

df.Text.head(5)
```

```
Out[5]: 0    bought several vitality canned dog food produc...
1    product arrived labelled lumbo halted peanuts....
2    connection around centuries. light, pillow cit...
3    looking secret ingredient robitussin believe f...
4    great staff great price. wide assortment mummy...
Name: Text, dtype: str
```

```
In [16]: # Create a new data frame "reviews" to perform exploratory data analysis upon_
reviews = df # Dropping null values
reviews.dropna(inplace=True) # The histogram reveals this dataset is highly un
reviews.Score.hist(bins=5,grid=False)
plt.show()
print(reviews.groupby('Score').count().Id)
```



```
Score
1      8
2      4
3      9
4     12
5     67
Name: Id, dtype: int64
```

```
In [18]: #To make it balanced data, we sampled each score by the lowest n-count from above.
score_1= reviews[reviews['Score'] ==1].sample(n=4)
score_2= reviews[reviews['Score'] ==2].sample(n=4)
score_3= reviews[reviews['Score'] ==3].sample(n=4)
score_4= reviews[reviews['Score'] ==4].sample(n=4)
score_5= reviews[reviews['Score'] ==5].sample(n=4)
#Here we create a 'balanced' dataset.
reviews_sample= pd.concat([score_1,score_2,score_3,score_4,score_5],axis=0)
reviews_sample.reset_index(drop=True,inplace=True)
#Printing count by 'Score' to check dataset is now balanced.
print(reviews_sample.groupby('Score').count().Id)
```

```
Score
1      4
2      4
3      4
4      4
5      4
Name: Id, dtype: int64
```

VADER Sentiment Analysis

```
In [6]: !pip install vaderSentiment
```

```
Requirement already satisfied: vaderSentiment in c:\users\vipul\anaconda3\lib\site-packages (3.3.2)
Requirement already satisfied: requests in c:\users\vipul\anaconda3\lib\site-packages (from vaderSentiment) (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\vipul\anaconda3\lib\site-packages (from requests->vaderSentiment) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\vipul\anaconda3\lib\site-packages (from requests->vaderSentiment) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\vipul\anaconda3\lib\site-packages (from requests->vaderSentiment) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\vipul\anaconda3\lib\site-packages (from requests->vaderSentiment) (2026.6.17)
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\lib\site-packages)
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\lib\site-packages)
```

```
In [7]: from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
        analyzer = SentimentIntensityAnalyzer()
```

```
# Generating sentiment for all the sentence present in the dataset
emptyline=[]
for row in df['Text']:
    vs=analyzer.polarity_scores(row)
    emptyline.append(vs)
```

```
In [8]: #Creating new dataframe with sentiments
        df_sentiments=pd.DataFrame(emptyline)
        df_sentiments.head(5)
```

```
Out[8]:
```

	neg	neu	pos	compound
0	0.000	0.517	0.483	0.9413
1	0.258	0.644	0.099	-0.5719
2	0.126	0.623	0.250	0.7880
3	0.000	0.868	0.132	0.4404
4	0.000	0.455	0.545	0.9186

Compound score is the metric that calculates the sentiment score(-1 to 1)

- positive > 0.05
- neutral > -0.05 and <0.05
- negative <= -0.05

```
In [9]: df_c = pd.concat([df.reset_index(drop=True), df_sentiments], axis=1)
df_c.head(3)
```

```
Out[9]:
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	Help
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian		1
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa		0
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"		1

```
In [10]: # Convert scores into postive and negative sentiments using some threshold
df_c['Sentiment']=np.where(df_c['compound']>=0 , 'Positive', 'Negative')
df_c.head(3)
```

Out[10]:

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	Help
--	----	-----------	--------	-------------	----------------------	------

0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian		1
---	---	------------	----------------	------------	--	---

1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa		0
---	---	------------	----------------	--------	--	---

2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"		1
---	---	------------	---------------	--	--	---

In [11]: `df_c[['Text', 'Sentiment']]`

Out[11]:

	Text	Sentiment
0	bought several vitality canned dog food produc...	Positive
1	product arrived labelled lumbo halted peanuts....	Negative
2	connection around centuries. light, pillow cit...	Positive
3	looking secret ingredient robitussin believe f...	Positive
4	great staff great price. wide assortment mummy...	Positive
...	...	...
95	pleased natural balance dog food. dog issue do...	Positive
96	1-1/2 year old basenji/jack russell mix love d...	Positive
97	up experienced gallery form hotspot itching do...	Positive
98	english bulldog skin gallery summer got age 3....	Positive
99	fed golden retrieved hated it. eat it, did, ga...	Negative

100 rows × 2 columns

In [12]: `%pip uninstall -y matplotlib`

Note: you may need to restart the kernel to use updated packages.

WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\Lib\site-packages)

WARNING: Skipping matplotlib as it is not installed.

```
In [13]: %pip install --no-cache-dir matplotlib
```

Collecting matplotlib

Downloading matplotlib-3.11.1-cp313-cp313-win_amd64.whl.metadata (80 kB)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (1.3.1)
Requirement already satisfied: cycler>=0.10 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.28.2 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (4.55.3)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (1.4.8)
Requirement already satisfied: numpy>=1.25 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (2.4.6)
Requirement already satisfied: packaging>=20.0 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=9 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (11.1.0)
Requirement already satisfied: pyparsing>=3 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (3.2.0)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\vipul\anaconda3\lib\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in c:\users\vipul\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
Downloading matplotlib-3.11.1-cp313-cp313-win_amd64.whl (9.3 MB)

```
----- 0.0/9.3 MB ? eta -:-:-
----- 0.0/9.3 MB ? eta -:-:-
- - - - - 0.3/9.3 MB ? eta -:-:-
- - - - - 0.3/9.3 MB ? eta -:-:-
-- - - - - 0.5/9.3 MB 452.8 kB/s eta 0:00:20
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 0.8/9.3 MB 650.6 kB/s eta 0:00:14
--- - - - - 1.0/9.3 MB 295.8 kB/s eta 0:00:29
--- - - - - 1.0/9.3 MB 295.8 kB/s eta 0:00:29
--- - - - - 1.3/9.3 MB 348.1 kB/s eta 0:00:24
--- - - - - 1.6/9.3 MB 399.6 kB/s eta 0:00:20
--- - - - - 1.8/9.3 MB 449.5 kB/s eta 0:00:17
--- - - - - 2.1/9.3 MB 480.6 kB/s eta 0:00:16
--- - - - - 2.4/9.3 MB 525.8 kB/s eta 0:00:14
--- - - - - 2.4/9.3 MB 525.8 kB/s eta 0:00:14
--- - - - - 2.6/9.3 MB 543.0 kB/s eta 0:00:13
--- - - - - 2.6/9.3 MB 543.0 kB/s eta 0:00:13
--- - - - - 2.6/9.3 MB 543.0 kB/s eta 0:00:13
--- - - - - 2.9/9.3 MB 527.3 kB/s eta 0:00:13
--- - - - - 2.9/9.3 MB 527.3 kB/s eta 0:00:13
--- - - - - 2.9/9.3 MB 527.3 kB/s eta 0:00:13
--- - - - - 3.1/9.3 MB 514.2 kB/s eta 0:00:13
--- - - - - 3.4/9.3 MB 531.5 kB/s eta 0:00:12
--- - - - - 3.4/9.3 MB 531.5 kB/s eta 0:00:12
```


-----	3.7/9.3	MB	533.7	kB/s	eta	0:00:11
-----	3.7/9.3	MB	533.7	kB/s	eta	0:00:11
-----	3.9/9.3	MB	551.6	kB/s	eta	0:00:10
-----	4.2/9.3	MB	561.4	kB/s	eta	0:00:10
-----	4.2/9.3	MB	561.4	kB/s	eta	0:00:10
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.5/9.3	MB	572.3	kB/s	eta	0:00:09
-----	4.7/9.3	MB	454.8	kB/s	eta	0:00:11
-----	5.0/9.3	MB	461.6	kB/s	eta	0:00:10
-----	5.0/9.3	MB	461.6	kB/s	eta	0:00:10
-----	5.2/9.3	MB	469.4	kB/s	eta	0:00:09
-----	5.2/9.3	MB	469.4	kB/s	eta	0:00:09
-----	5.2/9.3	MB	469.4	kB/s	eta	0:00:09
-----	5.5/9.3	MB	472.2	kB/s	eta	0:00:09
-----	5.8/9.3	MB	481.1	kB/s	eta	0:00:08
-----	5.8/9.3	MB	481.1	kB/s	eta	0:00:08
-----	6.0/9.3	MB	489.1	kB/s	eta	0:00:07
-----	6.0/9.3	MB	489.1	kB/s	eta	0:00:07
-----	6.0/9.3	MB	489.1	kB/s	eta	0:00:07
-----	6.3/9.3	MB	484.8	kB/s	eta	0:00:07
-----	6.6/9.3	MB	497.7	kB/s	eta	0:00:06
-----	6.8/9.3	MB	509.5	kB/s	eta	0:00:05
-----	6.8/9.3	MB	509.5	kB/s	eta	0:00:05
-----	7.1/9.3	MB	512.6	kB/s	eta	0:00:05
-----	7.1/9.3	MB	512.6	kB/s	eta	0:00:05
-----	7.1/9.3	MB	512.6	kB/s	eta	0:00:05
-----	7.3/9.3	MB	508.9	kB/s	eta	0:00:04
-----	7.3/9.3	MB	508.9	kB/s	eta	0:00:04
-----	7.3/9.3	MB	508.9	kB/s	eta	0:00:04
-----	7.6/9.3	MB	503.0	kB/s	eta	0:00:04
-----	7.6/9.3	MB	503.0	kB/s	eta	0:00:04
-----	8.1/9.3	MB	521.9	kB/s	eta	0:00:03
-----	8.1/9.3	MB	521.9	kB/s	eta	0:00:03
-----	8.4/9.3	MB	528.8	kB/s	eta	0:00:02
-----	8.4/9.3	MB	528.8	kB/s	eta	0:00:02
-----	8.7/9.3	MB	529.2	kB/s	eta	0:00:02
-----	8.9/9.3	MB	535.8	kB/s	eta	0:00:01
-----	9.2/9.3	MB	546.9	kB/s	eta	0:00:01
-----	9.3/9.3	MB	550.1	kB/s	eta	0:00:00

Installing collected packages: matplotlib

```
Successfully installed matplotlib-3.11.1
```

Note: you may need to restart the kernel to use updated packages.

```
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\Li  
b\site-packages)  
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\Li  
b\site-packages)  
WARNING: Ignoring invalid distribution ~atplotlib (C:\Users\vipul\anaconda3\Li  
b\site-packages)
```

```
In [14]: import matplotlib  
import matplotlib.pyplot as plt  
  
print(matplotlib.__version__)  
print("Matplotlib working!")
```

```
Matplotlib is building the font cache; this may take a moment.  
3.11.1  
Matplotlib working!
```

```
In [ ]:
```